BAEP 557: Technology Commercialization
Organic Nanoparticles: Dental Application
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Commercialization Proposal Overview

1. Technology Background Research

The proposed technology uses nanoparticles, which are particles that typically range between 1 and 100 nanometers (nm); a nanometer is approximately one thousand millionth of a meter (10−9 m)[1].

In the early 2000's, the study and application of nanotechnology (the use of particles, structures, and molecules between 1 and 100 nm) rose, subsequently touching areas including physics, chemistry, biology, materials science, computer science, and engineering. The use of nanoparticles differs from other solutions in these scientific areas due to the nanoparticles' exceptionally small scale and ability to take advantage of scientific properties at that scale (for example, the ability to seamlessly pass through a cell wall).

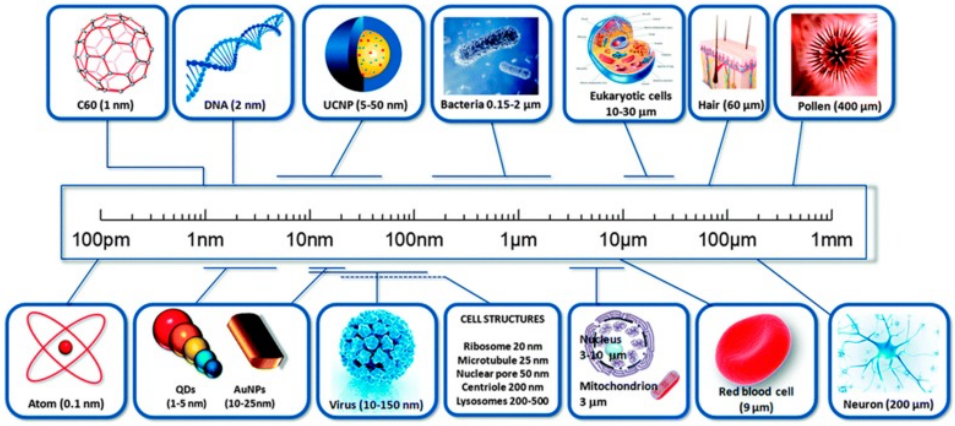


Figure 1. Nanomaterial comparisons by size.[1]
Source: National Center for Biotechnology Information

Since its introduction into the scientific community, the use and study of nanoparticles has contributed to numerous innovations and scientific advancements, outlined in **Figure 2** (continued on next page):

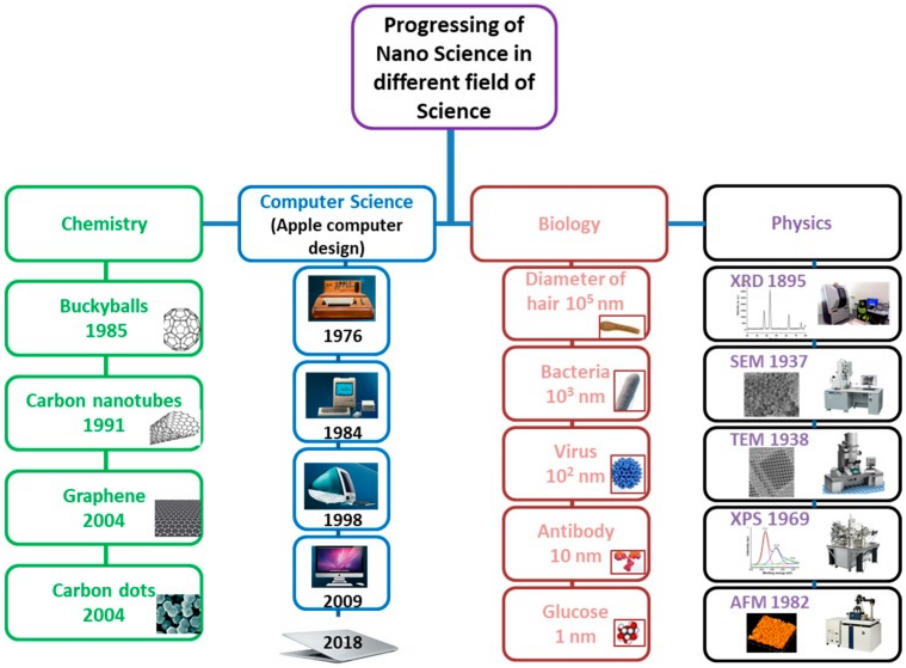


Figure 2. Application of Nanotechnology in Various Science-Related Fields. [1]
Source: National Center for Biotechnology Information

In the dental space, nanoparticles aim to combat the formation of biofilms produced by dental microbial germs through the use of various mechanisms. One such mechanism involves using NO (nitric oxide) and nanoparticles to penetrate the biofilm and eliminate the germs within.[2] Other methods include manipulating the size of nanoparticles by adjusting temperature, PH, and ion concentration, which disrupts the biofilm.[3] A third application and our area of focus involves nanoparticles to disrupt biofilms abrasively and routinely. NaOCl can also be used to dissolve the biofilm (Not the primary area of research).[4]

2. Research on Application Option

Dental plaque is a biofilm that consists of a wide variety of microorganisms and their by-products.[6] When uncalcified, this biofilm can be removed by routine oral hygiene aids—however, once calcified, their removal is difficult, even with the use of professional dental instruments, subsequently increasing the risk for disease. Dental clinicians historically report the eradication of biofilms as a significant challenge when managing their patients' risk for biofilm-associated diseases.[6]

Application Market

As a team, we are viewing the "application market" as the way we will deliver the existing nanoparticle technology (developed by AMI) to treat dental biofilms. Based on our preliminary research and feedback from Tom Lobl and Greg Zikos, our understanding of the effectiveness of nanoparticles is that they must be routinely introduced to a biofilm in order to penetrate and disrupt its structural integrity enough to prevent organisms from establishing residency in the biofilm. Currently, the most effective ways to treat and control biofilms are via mechanical methods such as tooth brushing and flossing, paired with routine care under a dental technician.[7] These methods rely heavily on the patient's dedication to a daily oral hygiene routine. With this in mind, we are targeting potential mediums that dental patients are already incorporating routinely into their dental hygiene practice, such as toothpaste, mouthwash, and whitening strips.

Customer Segment

Our team is focusing our primary "customer segment" on dental patients, with a secondary focus on dentists and dental hygienists. Based on our preliminary research and feedback from Tom Lobl and Greg Zikos, nanoparticles have the potential to help disrupt the biofilms most notably when used regularly and in addition to routine cleanings by a dental technician. Given that the regular cadence for routine cleaning at a dentist's office is typically every six months, our team is hypothesizing that a more feasible means for nanoparticles would be through an application suited for daily/regular use by the patient. As such, we are targeting the patient as the primary customer. We recognize that the dentist and/or dental hygienist plays a role in the education, treatment, and reinforcement of healthy habits in patients and therefore our team would like to target them as a secondary stakeholder market.

There are several options currently available for patients to manage biofilms today. Under the guidance of a dentist, dental hygienist, and/or other potential influencer, patients establish a daily oral health routine that typically involves brushing their teeth, flossing, and using a mouth rinse. The adherence to this daily routine varies significantly from patient to patient. In the United States alone, it has been reported that almost a third (33%) of Americans brush their teeth incorrectly.[8]

It is currently also recommended that the average dental patient visits a dentist's office every six months (for some patients, more or less) for routine cleaning and to have biofilms scraped off of their teeth. During a routine cleaning, dental technicians scrape the plaque off of teeth using dental instruments and treat conditions affecting the teeth and gums. Unfortunately, however, only 65.5% of adults aged 18-64 reported having a dental visit in the last 12 months, with that number even lower in rural areas.[9] Furthermore, having dentists scrape plaque does not prevent biofilms from forming. Only one layer of the biofilm is disrupted when plaque is scraped, and organisms in deeper layers of the biofilm can regenerate and form plaque again. There is currently no effective way to break down the biofilms that completely stops organisms from regenerating, and it would be impossible to keep end users away from being exposed to germs that form biofilms (they are everywhere, especially our food). A potential solution would comprise of a treatment that could break down biofilms faster than those biofilms are able to re-form and create plaque.

Cost Barriers

Similar to the traditional medical field, dentistry is associated with high costs that are split among the patient and the payer (insurance provider). Some solutions require the payer to include the solution on the formulary. Because of this dental cleanings can be expensive to the patient and some people may choose to avoid the professional dental procedures altogether. Additionally, routine oral hygiene necessities (such as toothpaste, floss, and mouthwash) are not covered by insurance and require a willingness to pay by the patient. Regulatory barriers dominate the dentistry environment and can delay the time to market and future profitability of new, more effective dental technologies.

Behavioral Barriers

While dentists and dental hygienists can make recommendations to patients, the burden of care relies on the patient to routinely manage his or her oral hygiene (i.e., brushing teeth daily). The burden also lies on the patient to schedule cleanings with a dental provider every six months. Oral hygiene and willingness to pay vary from patient to patient, and many patients may continue with their existing dental procedure even when a better

solution is introduced. Inadequate knowledge and fear of dental treatment can prevent many people from undergoing dental treatments.

3. Research Hypotheses

Our primary research hypotheses are as follows:

Patient (Primary Customer Segment)

1. How many dental patients follow a regular oral hygiene routine and why?
    a. What barriers exist that would prevent a dental patient from using a oral hygiene solution regularly?
    b. What incentivizes dental patients to use oral hygiene solutions on a regular basis?
    c. What daily oral hygiene they currently follow and how often they visit the dentists?
2. What types of patients care the most about their oral hygiene?
    a. Is it chronic patients?
    b. Who will be the most likely first adopters of dental nanotechnology?
3. Where do customers learn about their oral hygiene? Where do they learn about new products that are available?
    a. What role does the dentist/dental hygienist play in influencing dental patient's decisions? Are there other influencers/factors in play?
4. How many dental patients are looking for an alternate solution to existing dental treatments?
    a. How willing are they to switch or add products to their daily hygiene routine?
5. What is the patient's willingness to pay for better oral hygiene?
    a. Are there third variables that affect willingness to pay?

Dental Technician (Secondary Customer Segment)

1. What role does the dentist/dental hygienist play in influencing dental patients' decisions? Are there other influencers/factors in play?
2. How often does the average patient undergo a routine cleaning/plaque removal procedure in a year?
3. Will dentists be willing to promote a product that would eventually lead to a reduced need for dental visits?

1. Team Roles

While all team members will participate and contribute to all phases of the project, we will rotate leadership responsibilities for various elements of our commercialization project:

Team Member	Role	Key Responsibilities
Mihir Kavishwar	Product Manager	Linking features of the nanotechnology solution to benefits that successfully meet customer needs
Pranav Karthikeyan	Business Strategy and Development Manager	Research and identify new application opportunities for nanotechnology
Alex Weber	Commercial Leader	Synthesize and segment customer needs based on research insights, develop go-to-market strategy

Pranav Kulkarni	Research Leader	Ensure appropriate quantity and quality of interviews, designate notetakers and collect key insights
Mandi Wang	Financial Feasibility Leader	Assess financial viability / TAM, SAM, SOM. Evaluate the profitability of the propsed solution.

2. Next Steps & Information Needs

Milestone	Goal	Status	Next Step
Establish weekly Zoom meetings with Tom Lobl	Review existing research in the dental biofilms space	Complete	Next meeting scheduled for March 3
Contact Montana State University Center for Biofilm Engineering	Collect the latest research and trends in biofilms	In Progress	Establish team's availability during group meeting on Feb 28
Each team member to conduct five interviews with dental patients, and two interviews with dental providers	• 25 patient interviews and 10 dental provider interviews total • Source dental contacts from local practices and USC School of Dentistry	In Progress	Finalize Interview Guide and Identify Interviewees
Validate Application Options	Get feedback from Tom Lobl, Center of Biofilm Engineering and dentists on the potential application of the AMI nanoparticle solution	Not Started	Pause until customer interviews are complete
Review research materials from AMI	Deepen technical understanding of nanotechnology	In Progress	Review During Group Meeting on Feb 28

Works Cited

1. "Home - PMC - NCBI." *National Center for Biotechnology Information*, U.S. National Library of Medicine, https://www.ncbi.nlm.nih.gov/pmc/.
2. *Ferumoxytol Nanoparticles Target Biofilms Causing ... - ACS Publications*. https://pubs.acs.org/doi/10.1021/acs.nanolett.1c02702.
3. authors, All, et al. "Effect of Ph-Sensitive Nanoparticles on Inhibiting Oral Biofilms." *Taylor & Francis*, https://www.tandfonline.com/doi/full/10.1080/10717544.2022.2037788.
4. *The Use of Nanoparticles to Control Oral Biofilm Formation*. https://journals.sagepub.com/doi/10.1177/0022034510377794.
5. "Plaque-Busting Nanoparticles Could Help Fight Tooth Decay." *Science*, https://www.science.org/content/article/plaque-busting-nanoparticles-could-help-fight-tooth-decay.
6. Chandki, Rita, et al. "Biofilms: A Microbial Home." *Journal of Indian Society of Periodontology*, U.S. National Library of Medicine, Apr. 2011, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3183659/.
7. *A Third of Adults Brush Their Teeth Incorrectly - Dentistry Today*. https://www.dentistrytoday.com/a-third-of-adults-brush-their-teeth-incorrectly/.
8. "Products - Data Briefs - Number 427 - December 2021." *Centers for Disease Control and Prevention*, Centers for Disease Control and Prevention, 21 Dec. 2021, https://www.cdc.gov/nchs/products/databriefs/db427.htm.